

Ch 1 Sequence alignment Algorithms

1.1 Edit graph and alignment

1.2 Scoring alignments, Scoring Scheme

δ	-	A	C	G	T
-	0	0	0	0	0
A	0	1			
C	0		1		
G	0			1	
T	0				1

unit scoring scheme



SIMPLER

$$X = x_1 x_2 \dots x_m$$

$$Y = y_1 y_2 \dots y_n$$

$$x_i, y_j \in \Sigma = \{A, C, G, T\} \quad \begin{matrix} \delta(x_i, y_j) \\ \delta(x_i, -) \\ \delta(-, y_j) \end{matrix}$$

$\sim (-, \gamma)$

Ex: $X = \{ A T T A C G$

$Y = \{ A T A T C G$

Score = 4

optimal
score $\begin{cases} A T T A - C G \\ A - T A T C G \end{cases}$

Score = 5

Edit Graph

$G_{x,y}$ edge labelled directed graph

$G = (V, E)$

$V = \text{vertex set}$
 $E = \text{edge set}$

Vertices $(M+1) \times (N+1)$ starting from 0

i, j row i
column j

Edges

gap in γ

$1 \leq i \leq m$

$1 \leq j \leq n$

$(i-1, j) \xrightarrow{\begin{pmatrix} x_i \\ - \end{pmatrix}} (i, j)$ label of the edge

gap in x

$$1 \leq i \leq m$$

$$1 \leq j \leq n$$



alignment edge

$$1 \leq i \leq m$$

$$1 \leq j \leq n$$



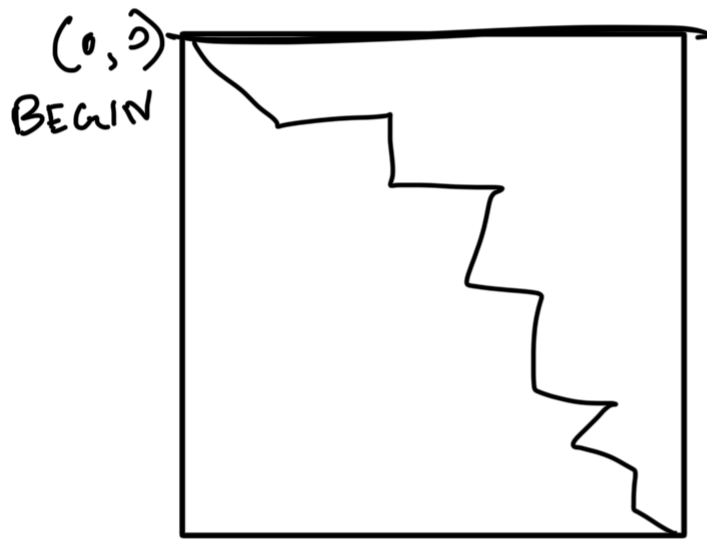
Conclusion

- Only 3 types of edges coming to (i, j)

Lemma/Theorem:

There is a 1-1 correspondence b/w directed paths from $(0, 0)$ to (m, n) and alignments

b/w X and Y



DAG

Directed
Acyclic
Graph } solves in
linear time

Define

Denote $S(i, j)$ the score of maximum path
from Begin to (i, j)

— Because there are only 3 edges directed into
vertex (i, j) it follows that any optimal(max)
path " P_{opt} " to (i, j) must use one of those
edges:

(1) P_{opt} first arrives at $(i-1, j)$

it follows gap in Y edge into (i, j)

(2) " " " " " (i, j-1)

... .. -- X edge into (i, j)

(3) P_{opt} gets to (i-1, j-1) pass through alignment edge gets to (i, j)

$$S(i, j) = \text{MAX} \begin{cases} S(i-1, j-1) + \delta(x_i, y_j) \\ \dots \dots \dots \\ \dots \dots \dots \end{cases}$$

Computing $S(i, j)$

We need to compute $S(i, j)$ such that at any time we have already computed the values of S needed by the _____

(at least one of the indices i and j smaller)

→ several ways to do:

* we will do column by column



initialised
first



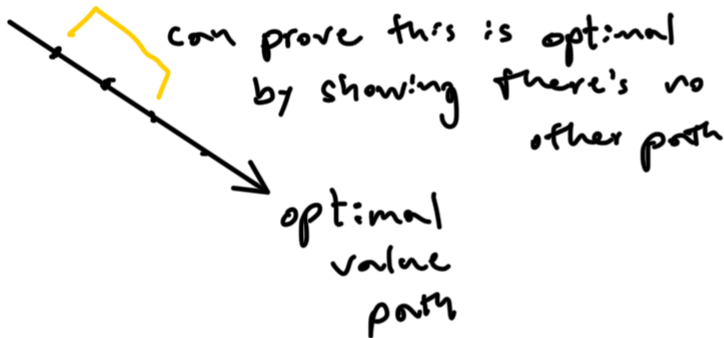
when we know one

we can find out the neighbouring nodes

Dynamic
programming
discipline



Graph G (directed)



Global alignment algorithm

$$S[0, 0] = 0$$

For $j=1$ to N DO

$$S[0, j] = S[0, j-1] + \delta(-, \tau_j)$$

For $i=1$ to N DO

$$S[i, 0] = S[i-1, 0] + \delta(x_i, -)$$

For $j=1$ to N DO

$$S[i, j] = \text{MAX} \begin{cases} S[\quad] + \delta(x_i, \tau_j) \\ S[\quad] + \delta(x_i, -) \\ S[\quad] + \delta(-, \tau_j) \end{cases}$$

PRINT ("Max", $S[M, N]$)

$$O(MN) \text{ if } M=N \rightarrow O(N^2)$$

Statistics

Conservation window

WWYIR

BABA

There are

ungapped
multiple
alignment

W	F	Y	V	R
W	Y	Y	V	R
W	Y	Y	T	R
W	F	Y	K	R
W	Y	Y	V	R

A	A	A	C
A	A	C	C
A	A	B	A
A	A	C	C
A	A	B	C

24 amino
acids observed
(6x4)

14 A
4 B
6 C

The observed proportions :

$$A \quad \frac{14}{24}$$

$$B \quad \frac{4}{24}$$

$$C \quad \frac{6}{24}$$

Pairs of letters aligned to each other:

$$4 \times \binom{6}{2} = 4 \times \frac{6 \times 5}{2} = 60 \text{ pairs}$$

Aligned pair	f observed
A to A	26/60
A B	8/60
A C	10/60

B	B	3/60
B	C	6/60
C	C	7/60

Observed $\rightarrow$ Expected f
 $\rightarrow$ need randomness

We will generate random sequences using the frequencies of aa ($A \frac{14}{24}, \dots$).

That is, we chose n random sequences with same f of aa given above, and make an alignment of them.

In a random alignment the EXPECTED proportion of pairs is

$$A \text{ to } A \quad \frac{14}{24} \times \frac{14}{24} \quad (\text{independence})$$

$$A \text{ to } B \quad 2 \left(n \text{ " } \right) \quad \begin{pmatrix} A \text{ to } B \\ B \text{ to } A \end{pmatrix}.$$

A to C

These fractions are now used to calculate the est. likelihood ratios:

$$2 \log_2 \left(\frac{\text{observed}}{\text{expected}} \right)$$

Aligned pair	obs f	exp f	log likelihood
A A	26/60	196/576	0.70
A B	:	112/576	-1.09
A C	\	168/576	-1.61
B B	/	16/576	1.70
B C	!	48/576	0.53
C C	7/60	46/576	0.83

A B C

A 1 -1 -2

B -1 2 1

C -2 1 2